ANNEXE 1. Extrait de quelques Transformées de Laplace usuelles

f(t)	F(P)
Impulsion de Dirac : $\delta(t)$	1
Échelon unitaire (fonction constante): u(t)=1	$\frac{1}{p}$
Rampe unitaire: t.u(t)	$\frac{1}{p^2}$
$\left(\frac{t^{n-1}}{(n-1)!}\right)u(t)$ n entier $n \geq 1$	$\frac{1}{p^n}$
$e^{-at}u(t)$	$\frac{1}{p+a}$
$te^{-at}u(t)$	$\frac{1}{(p+a)^2}$
$\left(e^{-at}\frac{t^{n-1}}{(n-1)!}\right)u(t)$ n entier $n \geq 1$	$\frac{1}{(p+a)^n}$
$\cos(\omega t)\ u(t)$	$\frac{p}{p^2+\omega^2}$
$e^{-at}\cos(\omega t)\ u(t)$	$\frac{p+a}{(p+a)^2+\omega^2}$
$\sin(\omega t)\,u(t)$	$\frac{\omega}{p^2+\omega^2}$
$e^{-at}\sin(\omega t)u(t)$	$\frac{\omega}{(p+a)^2+\omega^2}$
$\mathrm{ch}(\omega t)\ u(t)$	$\frac{p}{p^2-\omega^2}$
$sh(\omega t)\,u(t)$	$\frac{\omega}{p^2-\omega^2}$
$\left(\frac{e^{-t/\tau}}{\tau}\right)u(t)$	$\frac{1}{1+\tau p}$
$\left(1-e^{-t/\tau}\right)u(t)$	$\frac{1}{p(1+\tau p)}$
$\left(t-\tau+\tau e^{-t/\tau}\right)u(t)$	$\frac{1}{p^2(1+\tau p)}$
$\left(\frac{e^{-t/\tau_1}-e^{-t/\tau_2}}{\tau_1-\tau_2}\right)u(t)$	$\frac{1}{(1+\tau_1 p)(1+\tau_2 p)}$
$\left(1-\frac{\left(\tau_1 e^{-t/\tau_1}-\tau_2 e^{-t/\tau_2}\right)}{\tau_1-\tau_2}\right)u(t)$	$\frac{1}{p(1+\tau_1 p)(1+\tau_2 p)}$
$\left(t-(\tau_1+\tau_2)+\frac{\left(\tau_1^2 e^{-t/\tau_1}-\tau_2^2 e^{-t/\tau_2}\right)}{\tau_1-\tau_2}\right)u(t)$	$\frac{1}{p^2(1+\tau_1 p)(1+\tau_2 p)}$
$\left(\frac{(\tau-t)e^{-t/\tau}}{\tau^3}\right)u(t)$	$\frac{p}{(1+\tau p)^2}$
$\left(\frac{t}{\tau^2}e^{-t/\tau}\right)u(t)$	$\frac{1}{(1+\tau p)^2}$
$\left(1-\left(1+\frac{t}{\tau}\right)e^{-t/\tau}\right)\ u(t)$	$\frac{1}{p(1+\tau p)^2}$
$\left(t-2\tau+(t+2\tau)e^{-t/\tau}\right)u(t)$	$\frac{1}{p^2(1+\tau p)^2}$
$\left(\frac{{\omega_n}^2}{\sqrt{1-m^2}}e^{-m\omega_n t}sin(\omega_n\sqrt{1-m^2}t+\varphi)\right)u(t)$ $\varphi=\pi-\mathrm{ArcCos}(m)$	$\frac{p}{1+\frac{2m}{\omega_n}p+\left(\frac{p}{\omega_n}\right)^2}$
$\left(\frac{\omega_n}{\sqrt{1-m^2}}e^{-m\omega_n t}sin(\omega_n\sqrt{1-m^2}t)\right)u(t)$ $0<m<1$	$\frac{1}{1+\frac{2m}{\omega_n}p+\left(\frac{p}{\omega_n}\right)^2}$
$\left(1-\frac{1}{\sqrt{1-m^2}}e^{-m\omega_n t}sin(\omega_n\sqrt{1-m^2}t+\psi)\right)u(t)$ $\psi=\mathrm{ArcCos}(m)$	$\frac{1}{p\left(1+\frac{2m}{\omega_n}p+\left(\frac{p}{\omega_n}\right)^2\right)}$
$\left(t-\frac{2m}{\omega_n}-\frac{1}{\omega_n\sqrt{1-m^2}}e^{-m\omega_n t}sin(\omega_n\sqrt{1-m^2}t+2\psi)\right)u(t)$	$\frac{1}{p^2\left(1+\frac{2m}{\omega_n}p+\left(\frac{p}{\omega_n}\right)^2\right)}$